

When conditional variance harms KSIVI on the X-shaped target

Geometry, representation noise, and optimization boundaries

Reproducible investigation of the StatComp implementation

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Abstract

The empirical question is a comparison of training procedures, not a ranking of variational families. A conditional variance network contains the global variance model, and the population KSIVI objective depends on the marginal distribution rather than on its hierarchical representation. The X-shaped target has an exact representation with a global variance of $0.2I$ and a mixture of rank-one means. Conditional variance therefore introduces redundant directions without being necessary for this target. We prove that these directions can substantially increase conditional-score representation noise while leaving the marginal distribution unchanged. Independent GPU experiments separate this mechanism from initialization, kernel bandwidth differentiation, variance floors, annealing, and target geometry. The evidence supports representation noise as a major mechanism, amplified by the optimization schedule, but does not establish a universal inferiority theorem for conditional variance. All numerical outcomes, including failed explanations and counterexamples, are retained below.

1 What exactly is the phenomenon?

Write $\epsilon \sim N(0, I_2)$ and $u \sim N(0, I_2)$ independently. The two families implemented in this repository are

$$x = \mu_\phi(\epsilon) + \sigma_\phi(\epsilon) \odot u, \quad x = \mu_\phi(\epsilon) + \sigma_\phi \odot u.$$

Both means have two width-128 SiLU hidden layers. Conditional variances are

$$v_j(\epsilon) = \max\{10^{-4}, \text{softplus}(a_j(\epsilon))\}.$$

Global variances have the same softplus transform applied to two learnable scalars. The conditional model shares its hidden layers between mean and variance outputs. Changing `train.vi.var_lr` does not change its variance learning rate: the production optimizer creates a separate variance group only for models with a global `var_raw` parameter.

The canonical training policy is Adam (0.9, 0.999), initial learning rate 10^{-3} , batch size 128 in each of two independent batches, 50,000 updates, and StepLR with factor 0.9 every 1,000 updates. Target scores are multiplied by $\alpha_t = 0.1 + 0.9 \min(t/25000, 1)$. The Gaussian kernel and its empirical median bandwidth are differentiated. The original authors used a global variance vector in their experiments; their toy architecture and several optimization choices differ from this repository [1, 2].

The precise empirical comparison is the distribution over random seeds of the final approximation error under this policy. In the matched experiments, the initial mean weights are matched, the variance values agree to float32 roundoff, and the training noise streams are paired. A positive difference

$$\Delta_s = \text{SW}_2(q_{s,50000}^{\text{conditional}}, p) - \text{SW}_2(q_{s,50000}^{\text{global}}, p)$$

means worse conditional performance for seed s . The observed paired differences are: Seed 42: +2.380; Seed 43: +2.480; Seed 44: +0.391. Tables provide additional KL and KSD diagnostics; a discrepancy in one metric alone is not treated as proof.

This is not a statement that every conditional run fails, that every global run succeeds, or that conditional variance is inferior under another optimizer, estimator, architecture, or target. The statistical unit is an independently trained model, not a sample drawn from it. Three seeds support these concrete contrasts but do not locate a universal phase transition.

2 Population objective: the larger family is not intrinsically worse

Define $s_p = \nabla \log p$, $s_q = \nabla \log q_\phi$, and $s_c(x, \epsilon) = \nabla_x \log q_\phi(x | \epsilon)$. For the Gaussian conditional layer, $s_c = -u/\sigma_\phi(\epsilon)$. The executable code returns the *negative* conditional score u/σ , so its sign is correct. For an independently fixed kernel,

$$L_k(\phi) = \mathbb{E}[k(x, x') f(x, \epsilon)^T f(x', \epsilon')], \quad f = s_p - s_c.$$

The prime denotes an independent joint draw. Conditional integration gives the Fisher identity $\mathbb{E}[s_c | x] = s_q(x)$, hence

$$L_k(\phi) = \mathbb{E}_{q_\phi \otimes q_\phi} [k(x, x') (s_p(x) - s_q(x))^T (s_p(x') - s_q(x'))] = \text{KSD}_k^2(q_\phi, p).$$

This is the population identity underlying KSIVI [1]. It remains valid for a heteroscedastic conditional layer under the usual integrability and differentiation conditions. There is no additional population Fisher penalty for conditional variance and no inevitable population bias caused by conditioning the variance. The conditional network can realize the global network by setting its variance-output weights to zero and choosing constant biases. Therefore

$$\inf_{\phi \in \Phi_{\text{conditional}}} L_k(\phi) \leq \inf_{\phi \in \Phi_{\text{global}}} L_k(\phi).$$

This inequality concerns global optimization of a fixed population objective; it says nothing about the outcome of finite stochastic Adam training.

For annealed scores the same calculation yields the KSD against $p_\alpha(x) = p(x)^\alpha / Z_\alpha$. Here $\alpha \geq 0.1$, so the target is proper. It is a moving objective until update 25,000, not simply noisy training on p throughout. In particular p^α is generally not the mixture of the individually tempered Gaussian components.

3 Exact target geometry and a representational boundary

The target is

$$p = \frac{1}{2}N(0, \Sigma_+) + \frac{1}{2}N(0, \Sigma_-), \quad \Sigma_\pm = \begin{pmatrix} 2 & \pm 1.8 \\ \pm 1.8 & 2 \end{pmatrix}.$$

Its eigenvalues are 3.8 and 0.2. Let $S \in \{-1, +1\}$ be equiprobable, let $T \sim N(0, 1)$ and $U \sim N(0, I_2)$ independently. Then

$$X = \sqrt{1.8}T(1, S)^T + \sqrt{0.2}U \tag{3.1}$$

has exactly density p . Conditional on S , the covariance is Σ_S . The global Gaussian layer supplies the narrow transverse width, while the mixing means supply the long arms. This is a distributional construction; a finite

continuous neural mean approximates the discrete branch switch. It is not a claim that a finite width-128 network represents the switch exactly.

Proposition 3.1 (Exact global-noise boundary). *For an unrestricted mixing distribution of means and independent Gaussian noise $N(0, D)$ with $D = \text{diag}(d_1, d_2) \geq 0$, an exact representation of p exists if and only if $\Sigma_{\pm} - D \geq 0$. Equivalently,*

$$d_1 \leq 2, \quad d_2 \leq 2, \quad (2 - d_1)(2 - d_2) \geq 3.24.$$

For isotropic noise $D = vI$, this is $0 \leq v \leq 0.2$.

Proof. Sufficiency follows by mixing $N(0, \Sigma_+ - D)$ and $N(0, \Sigma_- - D)$ and adding $N(0, D)$. Degenerate Gaussian mean components are allowed. For necessity, suppose a direction w satisfies $w^T D w > \min_s w^T \Sigma_s w$. The characteristic function of the mixing mean in this direction would be

$$\frac{1}{2} \sum_{s \in \{+, -\}} \exp\{-t^2 w^T (\Sigma_s - D) w / 2\}.$$

At least one term grows without bound as $|t| \rightarrow \infty$, contradicting the bound $|\varphi(t)| \leq 1$ for a characteristic function. The matrix and diagonal conditions follow. $\square$

Conditional variance offers no needed expressiveness for this target in the closure of flexible mean mixtures. The upper endpoint 0.2 also gives a useful interpretation to successful learned global variances near 0.2; their proximity to that value is not, by itself, a proof of exact recovery. For the correlation-controlled target with off-diagonals $\pm 2\rho$, the corresponding isotropic boundary is $2(1 - |\rho|)$: 2, 1, and 0.2 at $\rho = 0$, 0.5, and 0.9. Sharper arms require smaller component noise.

For diagnostics, $\mathbb{E}X = 0$, $\text{Cov } X = 2I$, and $\mathbb{E}\|X\|^4 = 44.96$. With rotated coordinates $A = (X_1 + X_2)/\sqrt{2}$ and $B = (X_1 - X_2)/\sqrt{2}$, $\mathbb{E}A^2 B^2 = 3.8 \times 0.2 = 0.76$. An isotropic $N(0, 2I)$ has the same covariance but cross-fourth moment 4. Covariance recovery alone therefore cannot validate the X-shaped geometry.

4 Why conditional scores can become noisy without changing the marginal

Proposition 4.1 (Residual representation noise). *Assume $q_{\phi} = p$, Gaussian conditionals with diagonal covariance $D(\epsilon)$, and finite second moments of the conditional score. Then*

$$\mathbb{E}\|s_p(X) - s_c(X, \epsilon)\|^2 = \mathbb{E} \text{tr } D(\epsilon)^{-1} - \mathbb{E}_p \|s_p(X)\|^2. \quad (4.1)$$

Proof. Given X , the mean conditional score equals $s_p(X)$. Expanding the squared residual and using that identity reduces the expression to $\mathbb{E}\|s_c\|^2 - \mathbb{E}\|s_p\|^2$. Given ϵ , the Gaussian score has second moment $\text{tr } D(\epsilon)^{-1}$. $\square$

At the marginal optimum, f has conditional mean zero but need not itself be zero. For Gaussian kernels $k(x, x) = 1$, the RKHS random feature $Y = k(X, \cdot) \otimes f$ has zero mean and covariance operator C with $\text{tr } C$ equal to the right side of (4.1). Two independent size- N averages give the loss estimate $\langle \bar{Y}, \bar{Y}' \rangle$, whose variance at this optimum is $\text{tr}(C^2)/N^2$. Away from the optimum, if $m = \mathbb{E}Y$, the variance is

$$\frac{2}{N} \langle m, Cm \rangle + \frac{1}{N^2} \text{tr}(C^2).$$

These are loss-variance identities, not a substitute for a gradient-variance bound. Increasing $\text{tr } C$ does not establish a monotone increase in every direction of C or in every gradient coordinate.

For each variance coordinate, Jensen gives $\mathbb{E}[1/v_j] \geq 1/\mathbb{E}[v_j]$, strictly if v_j is nonconstant. A moderate *mean* variance can thus conceal a large representation noise budget when a small subset of latent inputs approaches the floor. At $v = 10^{-4}$ the component score standard deviation is 100, compared with approximately 2.24 at $v = 0.2$. These scales do not automatically establish a gradient failure, because target and conditional score terms can cancel. We measure gradients separately.

For $\ell_j = \log \sigma_j$ and fixed mean, the pathwise residual derivative is

$$\frac{\partial f_i}{\partial \ell_j} = \alpha H_{p,ij}(X) \sigma_j u_j - \mathbb{1}\{i = j\} \frac{u_j}{\sigma_j}.$$

Network parameter derivatives multiply these terms by latent-dependent Jacobians. A conditional variance head supplies many such directions and shares the trunk with the mean head. Global variance restricts them to two coordinates and pools updates over all latent inputs. Small variances, large target curvature, and shared Jacobians can therefore interact. This is a mechanism, not a theorem that adding parameters always increases gradient variance under Adam.

4.1 A counterfactual with exactly equal marginal distributions

For any $v \in (0, 0.2]$, take the mixing means from $\frac{1}{2}N(0, \Sigma_+ - vI) + \frac{1}{2}N(0, \Sigma_- - vI)$. Adding $N(0, vI)$ gives exactly p . One may also randomize v between 0.02 and 0.2, since the conditional-on- v marginal is p in both cases. The random variance has mean 0.11, exactly matching a constant-0.11 representation. This isolates hierarchy and estimator noise from all approximation error and training path differences. This distributional simulation is not a finite neural-network representation; the training experiments provide the architecture-specific link.

The experiment differentiates a common global dilation $X_\theta = e^\theta X$, $\sigma_\theta = e^\theta \sigma$ at $\theta = 0$, with fixed $h = 0.75$. It uses 256 independent repetitions of two size-128 batches. The population loss and its first derivative are zero in every row. The numerical variances are:

Component variance	Conditional gradient variance	Stein gradient variance	$\mathbb{E}\ f\ ^2$
0.2	0.00253	0.00196	5.93
0.11	0.01002	0.00201	14.11
0.05	0.04286	0.00204	35.93
0.02	0.28042	0.00217	95.93
0.02/0.20 mixture	0.07267	0.00186	50.93
0.002	24.45727	0.00203	995.93

The constant-0.02 representation has about 110 times the directional conditional-gradient variance of the constant-0.2 representation. The heteroscedastic representation has about seven times that of its constant-0.11 comparator. The Stein-estimator variance is approximately 0.002 across these representations. These measured ratios are finite Monte Carlo estimates, not universal exponents or constants.

4.2 An analytic gradient-variance asymptotic, with no fitted coefficient

Proposition 4.2 (Marginal-preserving gradient-noise explosion). *In the constant- v construction above, for fixed N and fixed Gaussian bandwidth $h > 0$, let g_v be the conditional-score estimator's derivative under global dilation at zero. Then*

$$\lim_{v \downarrow 0} v^2 \text{Var}(g_v) = \frac{2}{N^2} \mathbb{E}_{X, X' \sim p} \left[k_h(X, X')^2 \left(2 + \frac{\|X - X'\|^2}{h^2} \right)^2 \right] > 0.$$

In contrast, the Stein-estimator dilation gradient has exactly the same distribution for every v in this construction, including the heterogeneous representation.

Proof. Write $X_i = M_{v,i} + \sqrt{v} U_i$ and $X'_j = M'_{v,j} + \sqrt{v} U'_j$, where M_v has the residual Gaussian-mixture law and U, U' are standard two-dimensional normals independent of the means. At zero dilation, $\partial_\theta k_h = -\|X - X'\|^2 k_h / h^2$ and $\partial_\theta f = H_p(X)X - U/\sqrt{v}$. The leading pair-gradient term is

$$-\frac{1}{v} k_h(X, X') \left(2 + \frac{\|X - X'\|^2}{h^2} \right) U^T U'.$$

Couple M_v using the same mixture label and Gaussian coordinates as $v \downarrow 0$, so $M_v \rightarrow M_0$ with $M_0 \sim p$. The target score and its derivatives have polynomial growth and these Gaussian mixtures have all polynomial moments. Consequently $v g_v$ converges in L^2 to the average of these leading terms with X, X' replaced by M_0, M'_0 . Terms involving one conditional score vanish after multiplication by v . Distinct pair summands have zero covariance even if they share one index, since the other centered Gaussian noise has conditional mean zero. Also $\mathbb{E}(U^T U')^2 = 2$. This gives the stated positive limit. Finally, the Stein dilation gradient is a deterministic function of X, X' , whose independent marginal laws are p for every representation. Its entire distribution is therefore representation invariant. $\square$

The constant can be evaluated by a Gaussian quadratic-form Laplace transform. The difference $X - X'$ is a half-half mixture of normals with covariance eigenvalues (7.6, 0.4) and (4, 4). For either covariance C , put $t = h^{-2}$, $M = \det(I + 2tC)^{-1/2}$, $A = \text{tr}[C(I + 2tC)^{-1}]$, and $B = \text{tr}[(C(I + 2tC)^{-1})^2]$. The expectation in the limit is the mixture average of

$$M\{4 + 4tA + t^2(A^2 + 2B)\}.$$

At $h = 0.75$, $N = 128$, the resulting asymptotic variance is $9.9824 \times 10^{-5}/v^2$. It predicts 0.2496 at $v = 0.02$, compared with the measured 0.2804; no coefficient was fitted to that measurement. This is a small- v asymptotic, not an assertion of equality at finite v . It proves a concrete way that small conditional variances can defeat a uniform gradient-noise guarantee even at an exact marginal optimum. The actual code has a positive 10^{-4} variance floor. The asymptotic therefore explains potentially very large finite variance constants; it does not assert that the actual finite-floor runs have infinite variance.

Corollary 4.3 (Noise can also contaminate mean updates). *If the same construction dilates only the means, $X_\theta = e^\theta M_v + \sqrt{v} U$, while holding variance fixed, the conditional-score gradient still has inverse-square variance growth:*

$$\lim_{v \downarrow 0} v^2 \text{Var}(g_v^{\text{mean}}) = \frac{2}{N^2} \mathbb{E}_{p \otimes p} \left[k_h(X, X')^2 \frac{\|X - X'\|^4}{h^4} \right] > 0.$$

The Stein-estimator mean-direction gradient has bounded second moment as $v \downarrow 0$ for this target.

Proof. The conditional score no longer differentiates directly, but the kernel derivative multiplying the two conditional scores has leading term $-k_h(M_0, M'_0)\|M_0 - M'_0\|^2 U^T U'/(h^2 v)$. The preceding L^2 and pair-covariance argument applies. The Stein derivative contains no inverse variance and is bounded in L^2 by polynomial Gaussian moments. $\square$

Dilating mean outputs is an available parameter direction: scale the mean rows and biases of the final linear layer, leaving the variance rows unchanged. Thus the mechanism is not confined to the variance-head gradient under the independently fixed bandwidth of this corollary. Detaching the entire kernel removes this leading mean-gradient term but changes the update away from the full KSD gradient; detaching only the bandwidth does not remove it. Full adaptive-median differentiation needs a separate calculation: uniform dilation of all points also dilates the median bandwidth, so that the Gaussian kernel itself is invariant. The fixed-bandwidth coefficient must not be applied to that adaptive, isotropic mean direction.

This cancellation is not generic across mean parameters. To see why, scale only the first mean output, an available final-layer row direction. In the $v \downarrow 0$ construction, put $\delta_{ij} = M_{0,i} - M'_{0,j}$, $D_{ij} = \|\delta_{ij}\|^2$. Let (a, b) be the unique median-distance pair and $r_* = \delta_{ab,1}^2/D_{ab}$. For the implemented median rule, $h^2 = D_{ab}/[2 \log(N+1)]$, and differentiating the kernel gives

$$\partial_\theta k_{ij} = -\frac{k_{ij}}{h^2}(\delta_{ij,1}^2 - D_{ij}r_*) = -A_{ij}.$$

The leading conditional-score pair-average gradient is therefore $-(N^2 v)^{-1} \sum_{ij} A_{ij} U_i^T U'_j$. Conditional on the means, its variance is $2(N^4 v^2)^{-1} \sum_{ij} A_{ij}^2$. The median pair has zero coefficient, but other pairs do not all have its directional ratio almost surely for this continuous two-dimensional target and $N = 128$. Thus adaptive bandwidth differentiation does not remove all inverse-variance mean-gradient terms. This leading-term calculation is distinct from the proved full fixed-bandwidth variance asymptotic above, and is not a universal bound on every network-gradient coordinate.

The experiment's reported residual second moments use the exact identity (4.1) with the target Fisher information estimated independently from 200,000 exact target draws. They should not be confused with the closed-form gradient-asymptotic coefficient, which requires no such estimate.

5 Equivalent population objective, alternative estimator

Conditional integration by parts yields the conventional Stein integrand

$$h_p(x, y) = k s_p(x)^T s_p(y) + s_p(x)^T \nabla_y k + s_p(y)^T \nabla_x k + \text{tr}(\nabla_x \nabla_y k).$$

For $k = \exp(-\|x - y\|^2/(2h^2))$ this is

$$k \left[s_p(x)^T s_p(y) + \frac{(s_p(x) - s_p(y))^T (x - y)}{h^2} + \frac{2}{h^2} - \frac{\|x - y\|^2}{h^4} \right].$$

It removes the explicit u/σ term while retaining the same fixed-kernel population KSD. It still differentiates the sampling map and requires target score derivatives. It is not claimed to be a universally optimal Rao–Blackwell estimator. The exact Rao–Blackwell random feature would replace s_c by $\mathbb{E}[s_c | X] = s_q(X)$, which is generally intractable; the law of total covariance explains why removing latent-label noise is attractive, but does not rank all integration-by-parts gradient estimators.

In the training comparison, both the Stein variant and the bandwidth-only control detach the median bandwidth, while retaining spatial kernel gradients. Their comparison separates the estimator change from bandwidth detachment. A batch-dependent bandwidth means neither finite-batch loss is automatically an unbiased estimate of a single fixed-bandwidth population KSD. In particular, integration by parts for a jointly data-dependent bandwidth would contain additional derivative terms. The equal-objective claim is deliberately restricted to independently fixed bandwidths. All held-out KSD and exact-marginal mechanism experiments obey that restriction. The additional fixed- $h = 0.75$ conditional-versus-Stein training contrast also removes the adaptive-bandwidth confound; it keeps the same conditional architecture, target schedule, initialization, and optimizer.

6 Empirical evidence and the boundaries actually tested

6.1 Three-seed, 50,000-update comparisons

Means (sample standard deviations across trained seeds) are reported below. Each comparison starts at variance $\text{softplus}(1) = 1.31326$. Mean-network weights and biases are copied from the same seeded global initialization. Diagnostic sampling restores CPU and CUDA RNG states. The fast update has a locally verified gradient equal to the production update.

Procedure	n	SW_2	$\text{KL}(p\ q)$	$\text{KSD}_{h=0.75}^2$
Conditional: canonical	3	1.965 (1.136)	1.161 (0.378)	0.0449 (0.0307)
Global: canonical	3	0.214 (0.130)	0.275 (0.233)	0.0025 (0.0018)
Conditional: Stein estimator	3	0.283 (0.172)	0.357 (0.298)	0.0038 (0.0011)
Global: Stein estimator	3	0.371 (0.018)	0.874 (0.182)	0.0039 (0.0009)
Conditional: variance floor 0.2	3	0.400 (0.024)	0.579 (0.029)	0.0057 (0.0015)
Conditional: no anneal, constant LR	3	1.276 (1.486)	0.600 (0.596)	0.0528 (0.0770)
Global: no anneal, constant LR	3	0.167 (0.022)	0.049 (0.022)	0.0013 (0.0008)
Conditional: detach bandwidth	3	1.360 (1.050)	1.004 (0.420)	0.0193 (0.0115)
Conditional: fixed $h=0.75$	3	0.589 (0.168)	0.693 (0.277)	0.0061 (0.0016)
Conditional: fixed h , Stein	3	0.565 (0.188)	0.684 (0.220)	0.0104 (0.0067)

The canonical gap appears in all three paired seeds: mean SW_2 is 1.965 versus 0.214, and mean forward KL is 1.161 versus 0.275. Raising the conditional variance floor to 0.2 improves each canonical conditional seed, with mean SW_2 0.400. The adaptive-bandwidth Stein conditional variant also improves each canonical conditional seed, with mean SW_2 0.283; bandwidth detachment alone leaves mean SW_2 1.360.

The fixed-bandwidth comparison sets an important boundary: with $h = 0.75$, Stein improves SW_2 for seeds 42 and 43 but worsens it for seed 44. Its mean forward KL is nearly unchanged and mean held-out KSD is worse. The adaptive-bandwidth estimator improvement therefore does not establish a universal benefit from changing estimators, even when the population objectives agree. A successful Stein conditional run also retains a large inverse-variance tail: small component variance does not itself imply poor marginal approximation. It makes the conditional-score estimator noisy; the optimizer and kernel policy determine how that noise affects learning.

The variance-floor intervention restricts every conditional variance to at least 0.2, so the exact construction (3.1) remains available in the flexible-mean closure. It also changes optimization geometry and creates

zero gradients through clamped variance outputs. It is therefore a useful intervention but not a pure measurement of stochastic noise.

The no-annealing, constant-learning-rate comparison changes both the target path and adaptation budget. It tests a boundary of the observed phenomenon; it does not identify which of the two changes is responsible. The separate one-factor screening controls are retained in the appendix.

6.2 Historical canonical runs and initialization controls

We audited 334 saved checkpoints. One historical canonical result per family, seed, and initialization is retained in the table below; repeated runs using the same seed are not treated as independent seeds. These are historical comparisons, not matched-mean randomizations: changing the output layer shape can change initial mean biases under the same seed.

Family	Variance initialization	Seed	SW ₂	KL($p\ q$)	$\mathbb{E}[1/v]$
Global	Default	44	0.138	0.085	5.0
Global	Default	45	0.118	0.084	5.0
Global	Default	46	0.128	0.033	5.0
Conditional	Matched variance	42	0.518	0.463	29.4
Conditional	Matched variance	43	1.765	1.296	189.3
Conditional	Matched variance	44	0.705	0.962	8.2
Conditional	Default	42	4.215	1.366	2392.7
Conditional	Default	43	1.444	0.991	628.8
Conditional	Default	44	0.385	0.432	13.9
Global	Default	43	0.318	0.643	5.1
Conditional	Default	45	1.632	1.178	623.7
Conditional	Default	46	0.447	0.609	10.4

Matching only the initial variance in the historical conditional model does not guarantee recovery. The newer paired comparisons also match the mean function, which addresses this remaining initialization confound.

6.3 Broad screening and counterexamples

The screening study varies batch size, variance floors, network sharing, annealing duration, learning-rate decay, kernel bandwidth differentiation, estimator, and target correlation. These are 10,000-update, seed-42 results. Annealed runs still have $\alpha = 0.46$ unless a shorter warmup is specified; they are measured against the final target for consistency, so their absolute errors are not final-target convergence claims. At this horizon an unannealed conditional run beats its global counterpart, and the isotropic Gaussian control shows no comparable conditional disadvantage. Shared-trunk removal, larger batches, or bandwidth detachment alone do not establish a complete explanation of the original failure.

7 The schedule amplifies sensitivity; the published theorem has limits here

The canonical scheduler gives learning rate $10^{-3}0.9^{25} \approx 7.18 \times 10^{-5}$ when the final target first appears. Summing nominal learning rates gives approximately 9.282 before that point and 0.666 after it, through

update 50,000. Only about 6.7% of the total nominal step-size mass is spent on the final target. This is not an upper bound on Adam displacement, because adaptive normalization matters. It explains why more late iterations under the same decay need not repair an earlier basin.

Annealing initially favors a broader density; in the arms, transverse Gaussian variance scales approximately as $0.2/\alpha$. Both families undergo an early expansion, but latent-specific variance heads can develop different scales, small-variance tails, and branch imbalance during subsequent contraction. The temporal diagnostics support this interaction. Neither removing annealing alone nor changing decay alone is assumed to be a universal cure.

For the present target put $a = 2/0.76$ and $b = 1.8/0.76$. Up to a constant,

$$\log p(x, y) = -\frac{a}{2}(x^2 + y^2) + \log \cosh(bxy), \quad s_p = (-ax + by \tanh(bxy), -ay + bx \tanh(bxy)).$$

In particular,

$$\partial_{xx} \log p(0, R) = -a + b^2 R^2 \longrightarrow \infty.$$

Thus the global bounded-Hessian assumption used by the KSIVI optimization theorem is not literally satisfied even by this original toy target. Third derivatives are likewise unbounded along $x = c/(bR)$, $y = R$ with nonzero fixed c . A local or high-probability truncated analysis may be useful, but it is not the global theorem as stated. This fact applies to both families and is not itself a causal explanation of their ordering.

The published convergence result is to an approximate stationary point under smoothness, variance, and step-size assumptions, not to the global minimizer or a small KL approximation. Its variance constant also depends on parameterization smoothness and moments [1, 2]. Consequently the observed outcomes do not contradict an expressiveness or convergence claim made by that theorem. Spurious minima in KSD descent are an established possibility [3]; we do not use a higher-dimensional RBF non-convergence result as a proof of this two-dimensional failure.

8 What is established, and what remains an inference?

Established analytically: population representation invariance; nested variational families; the exact global Gaussian-noise boundary; the inverse-variance identity for representation noise; inverse-square gradient-variance growth in a marginal-preserving counterfactual, including an available mean-parameter direction; and unbounded target curvature.

Established empirically in these experiments: the numerical seed comparisons and intervention outcomes in the tables; the gradient-noise increase at exactly equal marginals; and the independent held-out accuracy and fixed-bandwidth KSD measurements. Each trained-model directional gradient comparison uses a common dilation and a fixed independent bandwidth.

Supported interpretation: redundant conditional scale directions can make the conditional-score stochastic estimator much noisier, particularly through a small-variance tail. The global model regularizes these directions and has a geometry well suited to this target. Kernel and optimizer choices influence which basin is reached; the annealing/decay schedule leaves little late adaptation. This is a combined explanation, not an assertion that one scalar diagnostic determines every run.

Not established: an exact critical correlation, a universal ranking for all kernels or parameterizations, a decomposition of Adam’s basin selection into separate causal percentages, or a guarantee that a particular variance floor or estimator works on all targets. The separate-network and correlation tests have one seed and a short horizon. The largest changes in KL across nested Monte Carlo mixture sizes are reported below, so uncertainty from density evaluation is not hidden.

For a KSIVI implementation intended for this target, a global variance layer is a justified first choice. If conditional variance is needed elsewhere, monitor lower variance quantiles, $\mathbb{E}[1/v]$, floor occupancy, and independent accuracy metrics; explicitly test estimator and schedule choices. The successful intervention in this study is evidence for a mechanism, rather than a blanket recommendation to replace the original KSIVI estimator.

9 Reproduction and measurement

All new experiments ran on the server specified by the reloaded `AGENTS.md`. The host was `connect.nmb1.seetacloud.com` (SSH port 37874), with an RTX 4090, PyTorch 2.9.0+cu126, and conda environment `ruivi`. Jobs ran in `tmux`. Code was committed locally, pushed, and fetched into the requested branch's remote worktree. Report data and figures were generated remotely and returned through Git. Raw checkpoints, samples, logs, and status files are under

```
/root/ruivi/results/ksivi_variance_investigation_20261006
```

There are 53 completed new training runs. The report source and its figures are under the branch's

```
campaigns/ksivi_x_shaped_evolution_20261006/investigation
```

Generation code commit: `c30bbe2d6e66149bb809b095a2ac0e210ca39577`. Individual run manifests record their own source commits; scripts were extended during the campaign, and later launches use those recorded revisions rather than an assumed common commit. The governing optimizer/architecture definitions are in `investigate.py`; the alternative estimator is explicitly distinguished from a production KSIVI update.

SW_2 is the root mean squared one-dimensional quantile displacement over 128 evenly spaced directions and 10,000 samples per distribution; it is not the full two-dimensional Wasserstein distance. Final independent diagnostics also draw 50,000 samples for moments. Forward KL uses 4,096 exact target draws and the explicit Gaussian-mixture density integrated over 4,096, 8,192, and 16,384 independent latent draws. For final runs with a change larger than 0.015 from 8,192 to 16,384, this is increased to 65,536 and 262,144 latent draws. The table uses the largest evaluated mixture. Finite latent integration introduces bias in log density, particularly for narrow components; this is checked by nested mixture sizes. Evaluation errors are separate from between-training-seed variation. KSD uses conventional Stein integrands on 16 independent pairs of size-256 batches for each of $h = 0.5, 0.75, 1.5$. Cross-batch estimates can be slightly negative, especially near zero.

Maximum absolute forward-KL changes between the last two evaluated mixture sizes, across seeds within each confirmation group:

Procedure	Maximum change
Conditional: canonical	0.038
Global: canonical	0.004
Conditional: variance floor 0.2	0.004
Conditional: no anneal, constant LR	0.041
Global: no anneal, constant LR	0.003
Conditional: Stein estimator	0.013
Global: Stein estimator	0.002
Conditional: detach bandwidth	0.009
Conditional: fixed $h=0.75$	0.006
Conditional: fixed h , Stein	0.013

A Per-seed accuracy and trained-model gradient noise

The last two columns are variances over 128 independent repetitions of the same dilation direction, at fixed $h = 0.75$. Parameter counts do not change the definition of this common direction. These are local diagnostics of the final trained distributions, not replicated training outcomes.

Procedure	Seed	SW_2	$KL(p q)$	$\mathbb{E}[1/\nu]$	$\text{Var } g_c$	$\text{Var } g_s$
canonical_cond	42	2.692	0.997	3242.0	766.3783	0.0017
canonical_cond	43	2.546	1.593	1200.4	110.4632	0.0049
canonical_cond	44	0.656	0.893	30.2	0.1076	0.0032
canonical_global	42	0.312	0.423	5.1	0.0036	0.0024
canonical_global	43	0.066	0.006	5.1	0.0032	0.0022
canonical_global	44	0.265	0.395	5.1	0.0044	0.0021
floor02_cond	42	0.373	0.606	4.6	0.0026	0.0032
floor02_cond	43	0.409	0.547	4.6	0.0026	0.0028
floor02_cond	44	0.419	0.583	4.6	0.0020	0.0033
no_anneal_const_cond	42	2.932	1.195	4300.4	972.7919	0.0020
no_anneal_const_cond	43	0.058	0.003	6.3	0.0042	0.0022
no_anneal_const_cond	44	0.839	0.604	200.7	6.2891	0.0023
no_anneal_const_global	42	0.188	0.072	5.3	0.0032	0.0022
no_anneal_const_global	43	0.144	0.029	5.1	0.0023	0.0017
no_anneal_const_global	44	0.168	0.048	5.5	0.0027	0.0018
stein_cond	42	0.086	0.022	1842.8	384.3731	0.0021
stein_cond	43	0.403	0.590	514.5	74.4273	0.0024
stein_cond	44	0.361	0.458	2089.8	555.6733	0.0034
stein_global	42	0.383	1.051	5.0	0.0031	0.0028
stein_global	43	0.380	0.688	5.1	0.0032	0.0024
stein_global	44	0.351	0.884	5.1	0.0028	0.0025
detach_h_cond	42	1.219	1.057	133.7	3.3694	0.0034
detach_h_cond	43	2.473	1.394	400.9	9.5097	0.0028
detach_h_cond	44	0.387	0.560	7.0	0.0047	0.0029
fixed_h_cond	42	0.734	0.933	60.8	0.6399	0.0026
fixed_h_cond	43	0.629	0.389	8.7	0.0073	0.0029
fixed_h_cond	44	0.405	0.757	8.3	0.0064	0.0026
fixed_h_stein_cond	42	0.440	0.623	679.0	101.1022	0.0024
fixed_h_stein_cond	43	0.473	0.501	480.2	31.4295	0.0030
fixed_h_stein_cond	44	0.781	0.927	10.3	0.0214	0.0035

B All screening outcomes

The cross-fourth target value is 0.76 for $\rho = 0.9$, 3 for $\rho = 0.5$, and 4 for $\rho = 0$. Large cross-fourth values can reflect escaped tails; values alone do not measure all aspects of approximation quality.

Screening procedure	SW_2	$\mathbb{E}[1/\nu]$	$\mathbb{E}[A^2 B^2]$
batch512_cond	1.193	5.2	7.173
batch512_global	0.714	2.7	3.775
canonical_cond	1.222	4.1	6.922
canonical_global	1.035	2.3	4.901
constant_lr_cond	1.353	4.8	7.825
constant_lr_global	0.874	2.4	3.921
detach_h_cond	2.046	83.0	11.684
detach_h_global	1.003	2.3	4.458
fast_warmup_cond	1.160	673.1	10.974
fast_warmup_global	0.505	5.1	1.016
floor002_cond	1.623	5.5	9.578
floor02_cond	1.039	3.3	5.458
gaussian_cond	0.668	0.2	18.891
gaussian_global	0.704	0.7	19.856
no_anneal_cond	0.534	8.2	0.901
no_anneal_const_cond	0.542	10.2	0.937
no_anneal_const_global	0.533	4.9	1.033
no_anneal_global	1.138	2.8	3.298
rho05_cond	0.832	1.2	22.785
rho05_global	0.801	0.7	15.090
separate_cond	1.320	1.8	6.848
stein_cond	1.013	1481.4	4.684
stein_global	1.094	2.2	5.005

C Artifact guide

`report.pdf` contains this mathematical report, all numerical tables, and the scientific figure appendix. `evidence_figures.pdf` is the companion figure booklet, with a separate final-sample page for each seed. `new_metrics.csv`, `historical_metrics.csv`, and `trajectories.csv` expose numerical outcomes without selecting favorable seeds. JSON files retain evaluation standard errors, nested density sizes, run source commits, and probe parameters. The LaTeX source of this report is `report.tex`; its vector figure assets are in `figures/`. Both PDFs are compiled from LaTeX. The recorded compiler versions, input hashes, and output hashes are in `typesetting.json`.

D Scientific figures

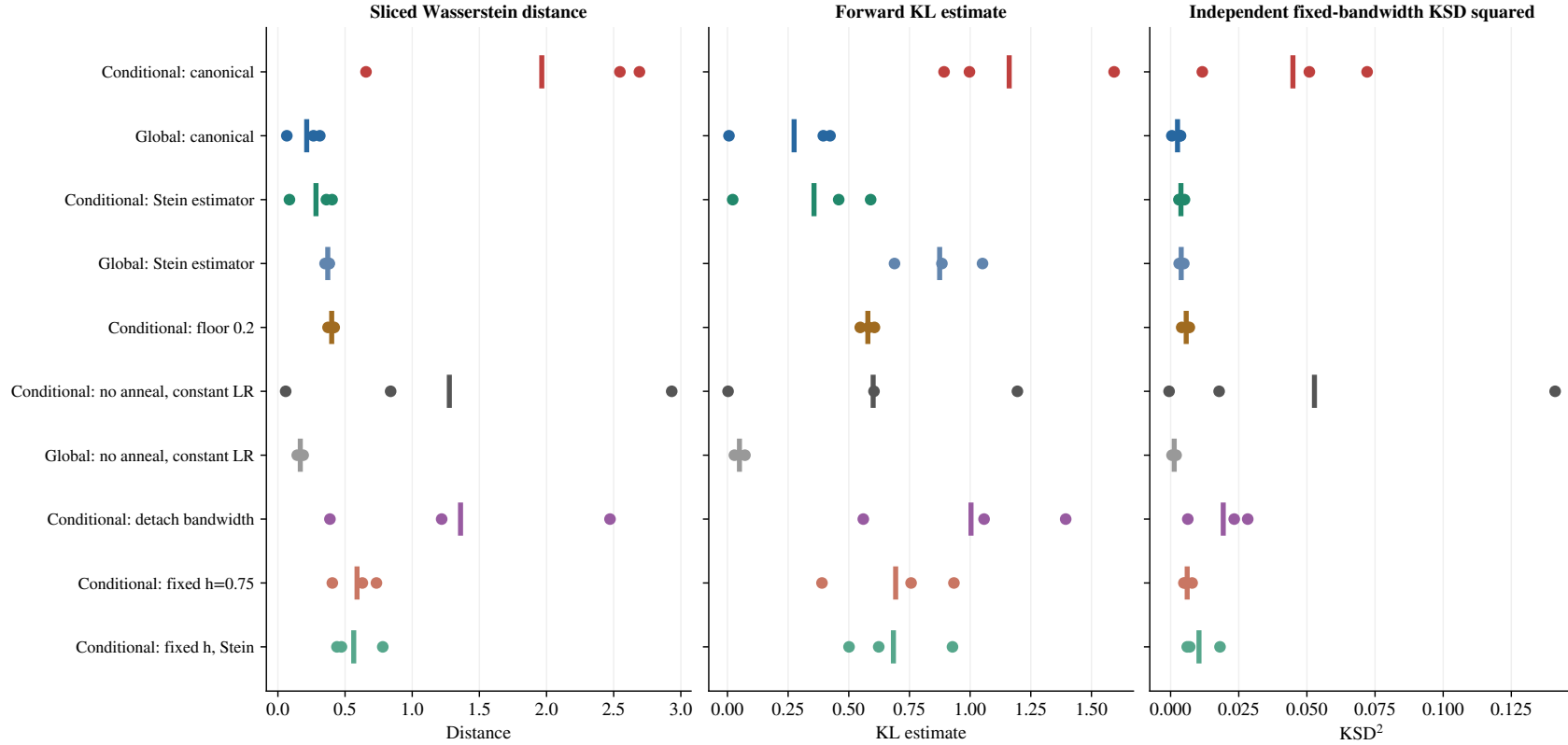


Figure 1. Matched comparisons after 50,000 updates. Dots are independently trained seeds 42–44; vertical strokes mark means. Initial mean weights, variance values, and training noise are paired. Forward KL uses the largest evaluated latent mixture; held-out KSD uses an independent fixed bandwidth of 0.75.

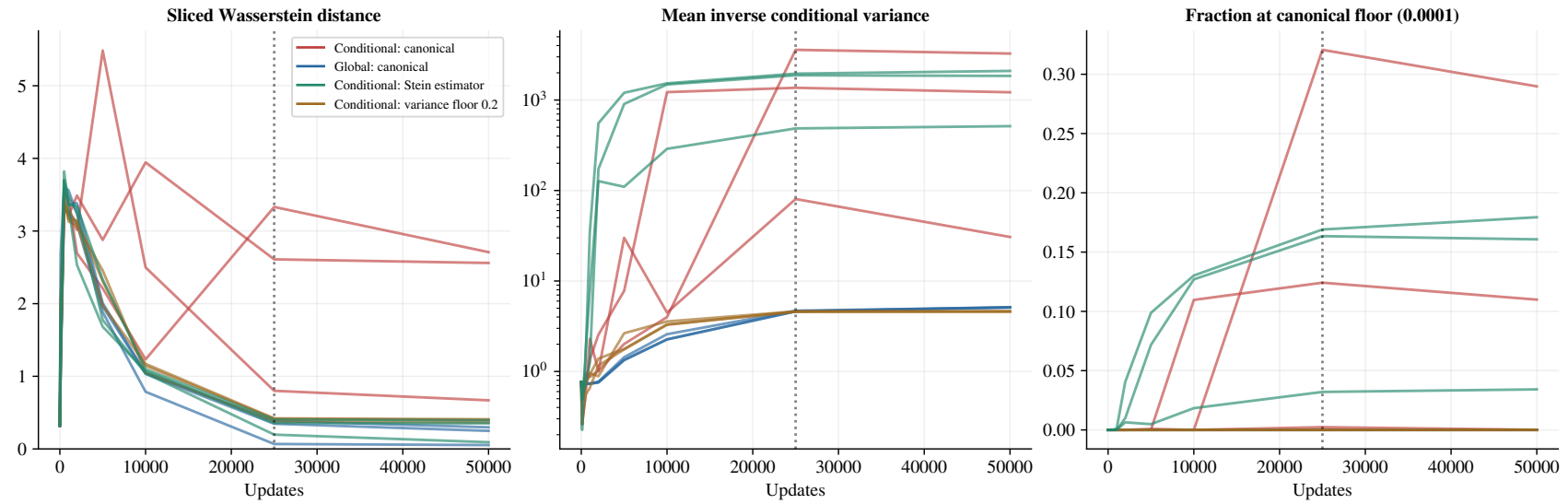


Figure 2. Accuracy and the small-variance tail across three trajectories per procedure. The dotted line marks completion of target annealing at update 25,000. Floor occupancy uses the common threshold 0.00010001; diagnostic draws preserve training RNG.

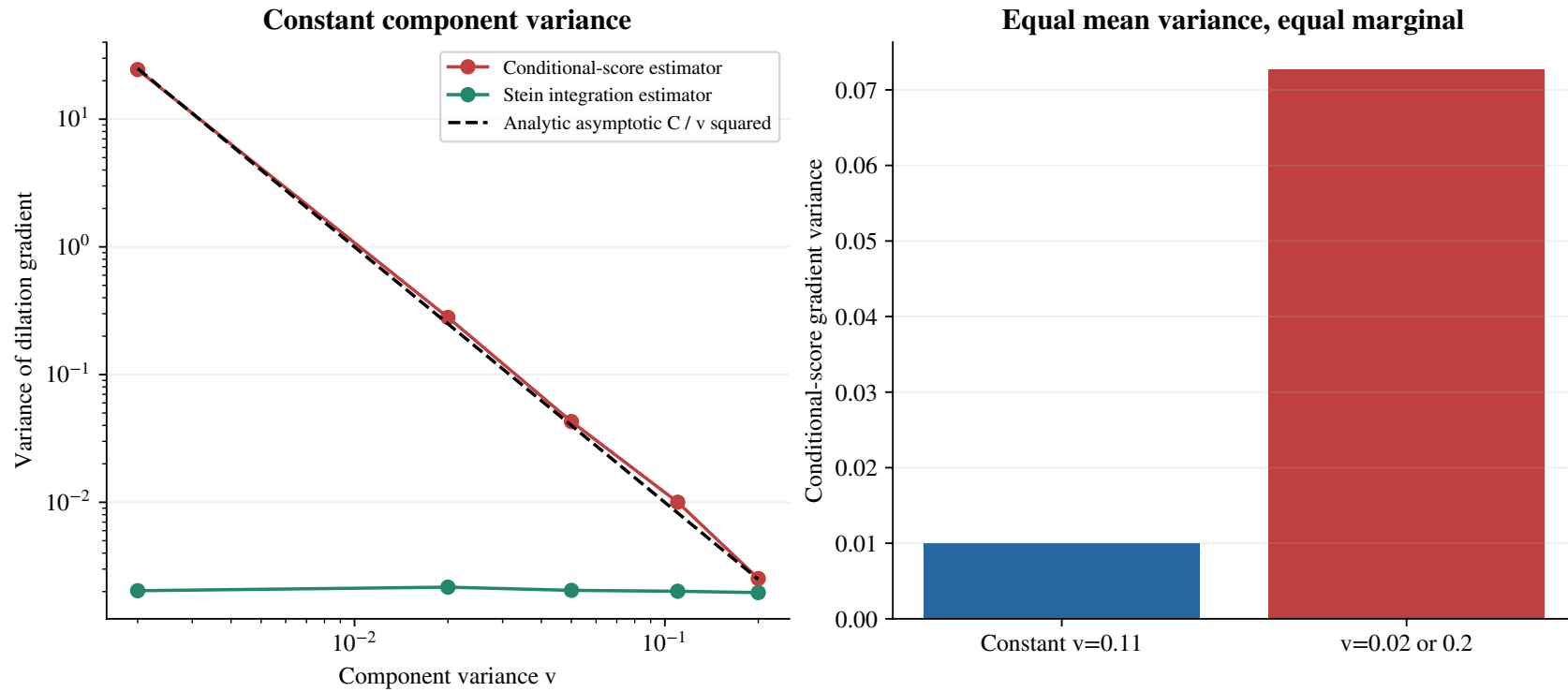


Figure 3. Representation noise at exactly equal marginal distributions. Each measurement uses 256 independent repetitions and two batches of 128 at fixed bandwidth 0.75. The dashed curve is the analytic small-variance prediction, with no fitted coefficient. The right panel holds both the marginal and mean component variance 0.11 fixed.

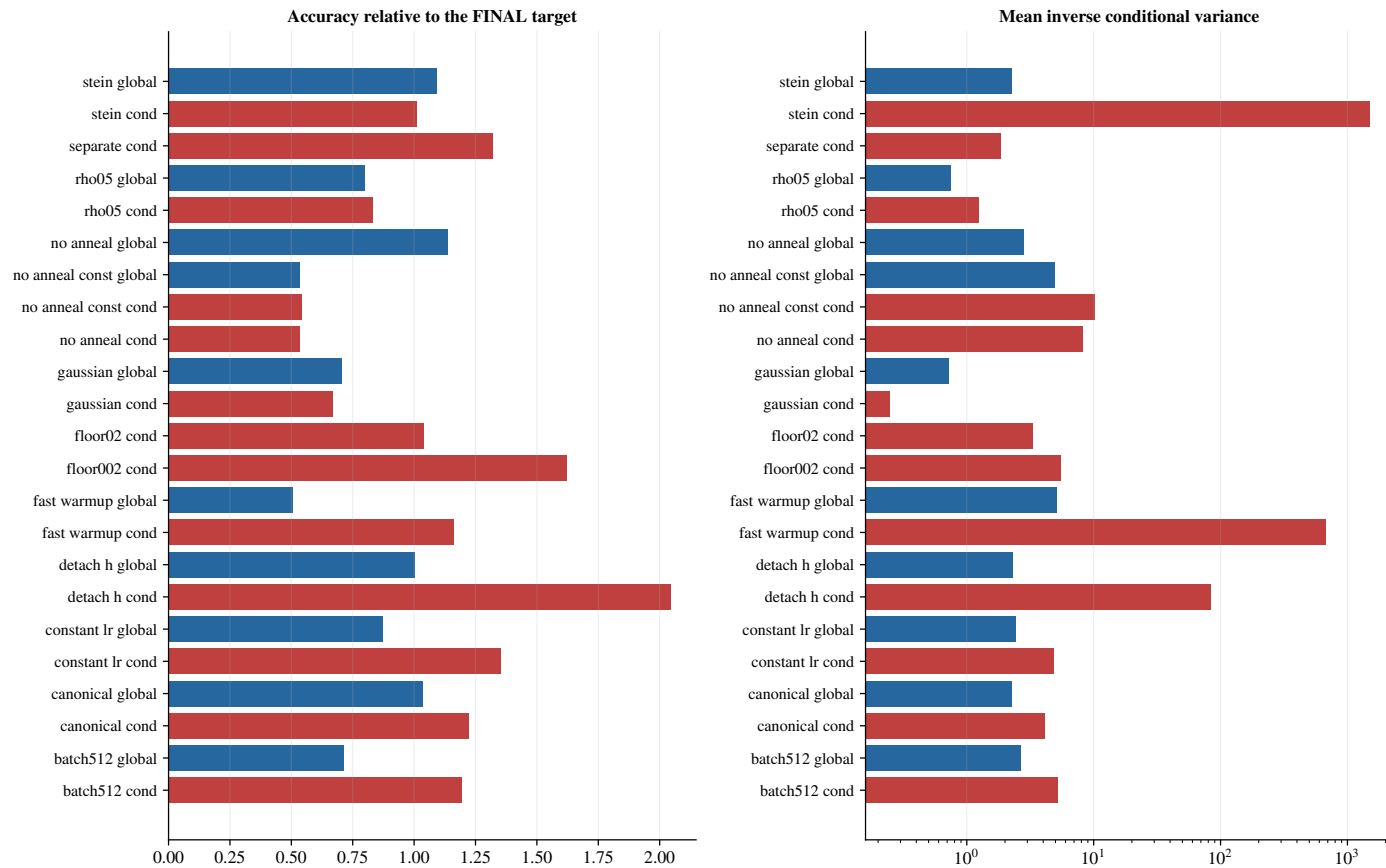


Figure 4. All 23 screening outcomes at 10,000 updates, seed 42. Blue denotes global variance; red denotes conditional variance. Annealed runs still have a score multiplier of 0.46 unless warmup is shortened. Distances are measured against the final target and do not establish final-target convergence.

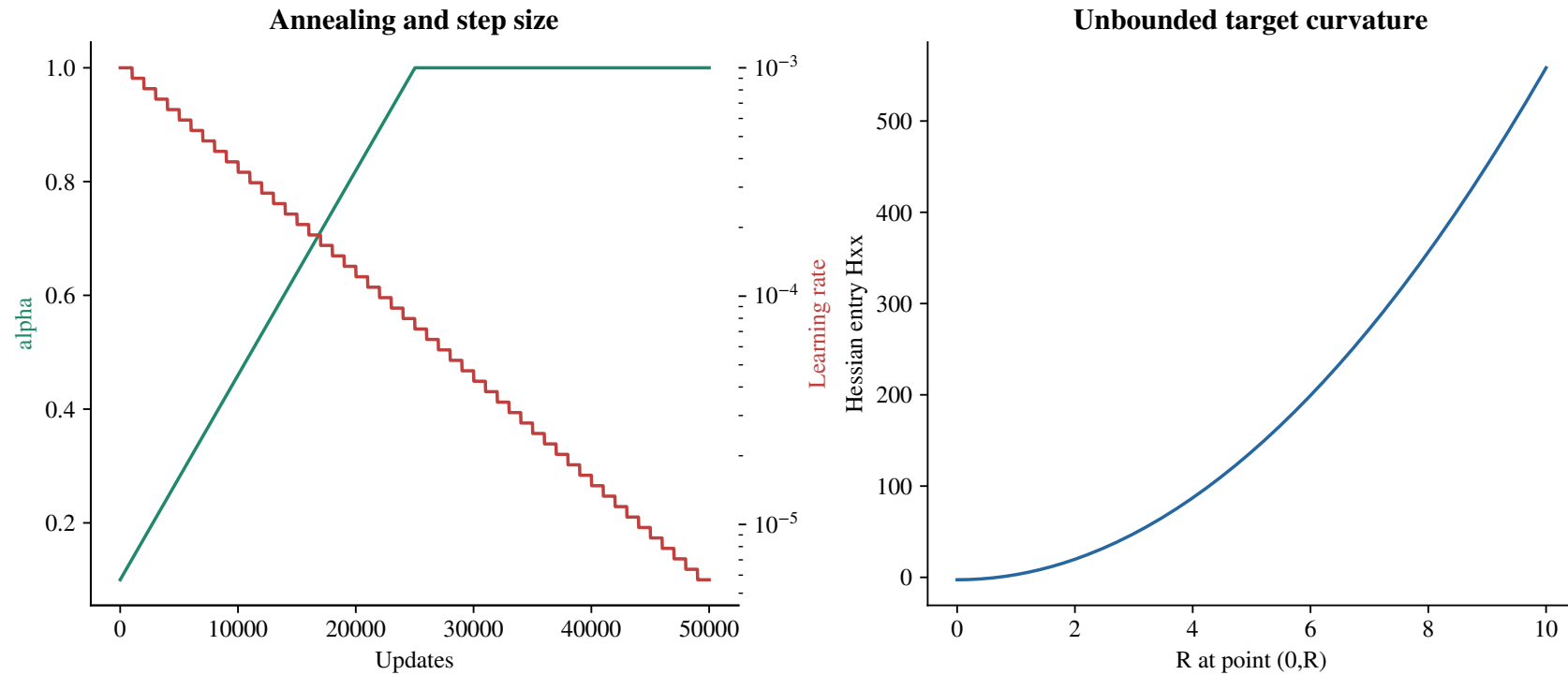


Figure 5. Target path, learning-rate decay, and target curvature. The score multiplier reaches one after the learning rate has fallen to approximately 0.0000718. The diverging Hessian entry violates the global bounded-Hessian assumption for both variational families; this fact alone does not explain their ordering.

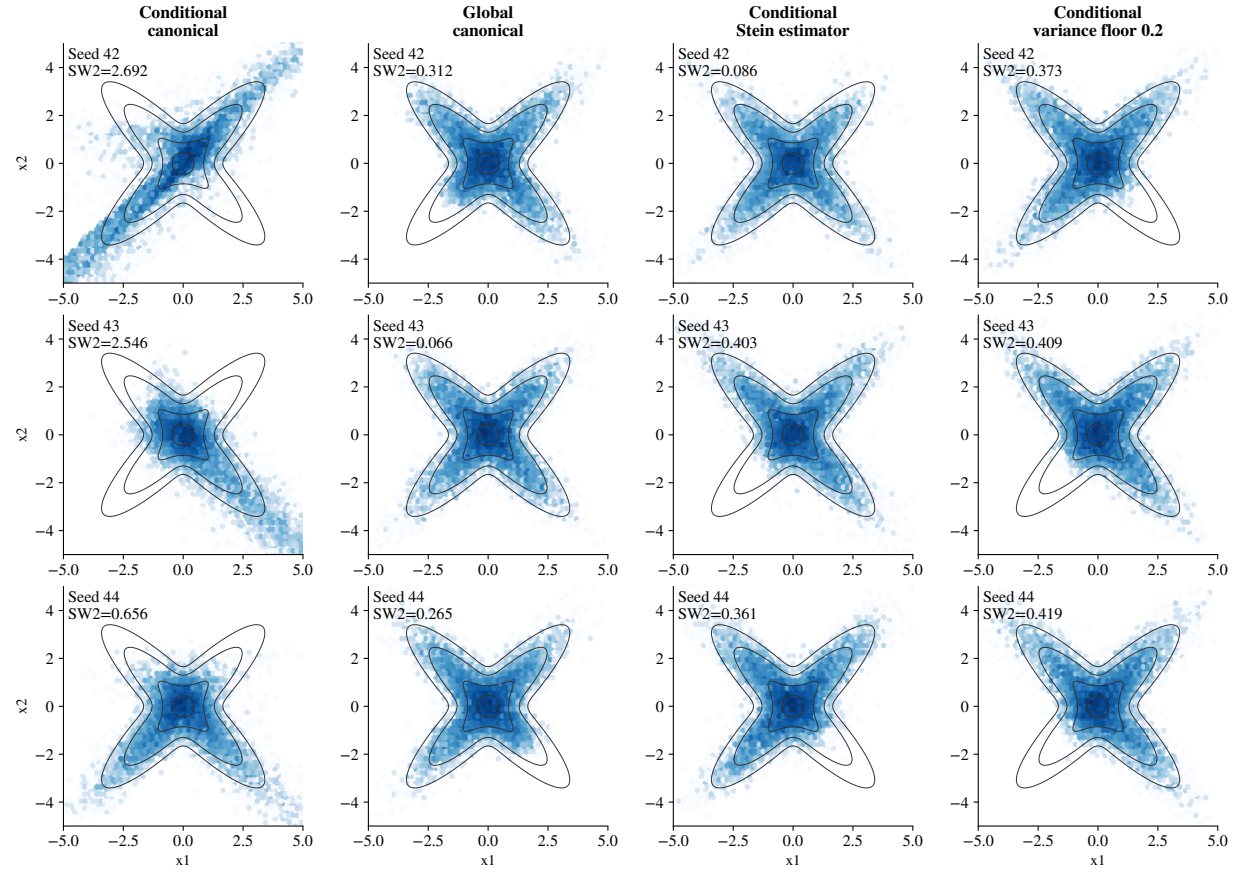


Figure 6. Final samples for seeds 42–44, with exact target contours and a common viewing window. All samples contribute to numerical metrics, including tails outside the plot. The separate sample pages in the companion booklet provide larger panels for each seed.

References

- [1] Z. Cheng, L. Yu, T. Xie, S. Zhang and C. Zhang (2024). Kernel Semi-Implicit Variational Inference. ICML, PMLR 235:8248–8269. <https://proceedings.mlr.press/v235/cheng24l.html>.
- [2] L. Yu, Z. Cheng, S. Zhang and C. Zhang (2026). A Kernel Approach for Semi-implicit Variational Inference. <https://arxiv.org/abs/2601.12023>. Appendix A.4 documents global variance in the experiments; Appendix C.1 gives toy-model settings.
- [3] A. Korba, P.-C. Aubin-Frankowski, S. Majewski and P. Ablin (2021). Kernel Stein Discrepancy Descent. ICML, PMLR 139:5719–5730. <https://proceedings.mlr.press/v139/korba21a.html>.
- [4] Original KSIVI implementation: <https://github.com/longinYu/KSIVI>.